MIL-F-16377/26A(SH)
9 May 1983
SUPERSEDING
MIL-F-16377/26(SHIPS)
30 November 1970

MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; INCANDESCENT, GENERAL LIGHTING

110 WATTS, 120 VOLTS, WATERTIGHT SYMBOLS 112, 113, and 113.1

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

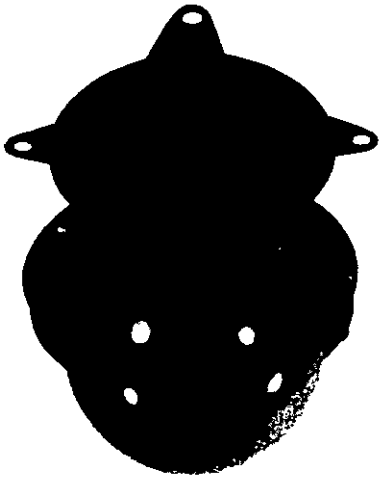


FIGURE 1. Fixtures, incandescent, Symbols 112 and 113.

NOTE:

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

FSC 6210

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.

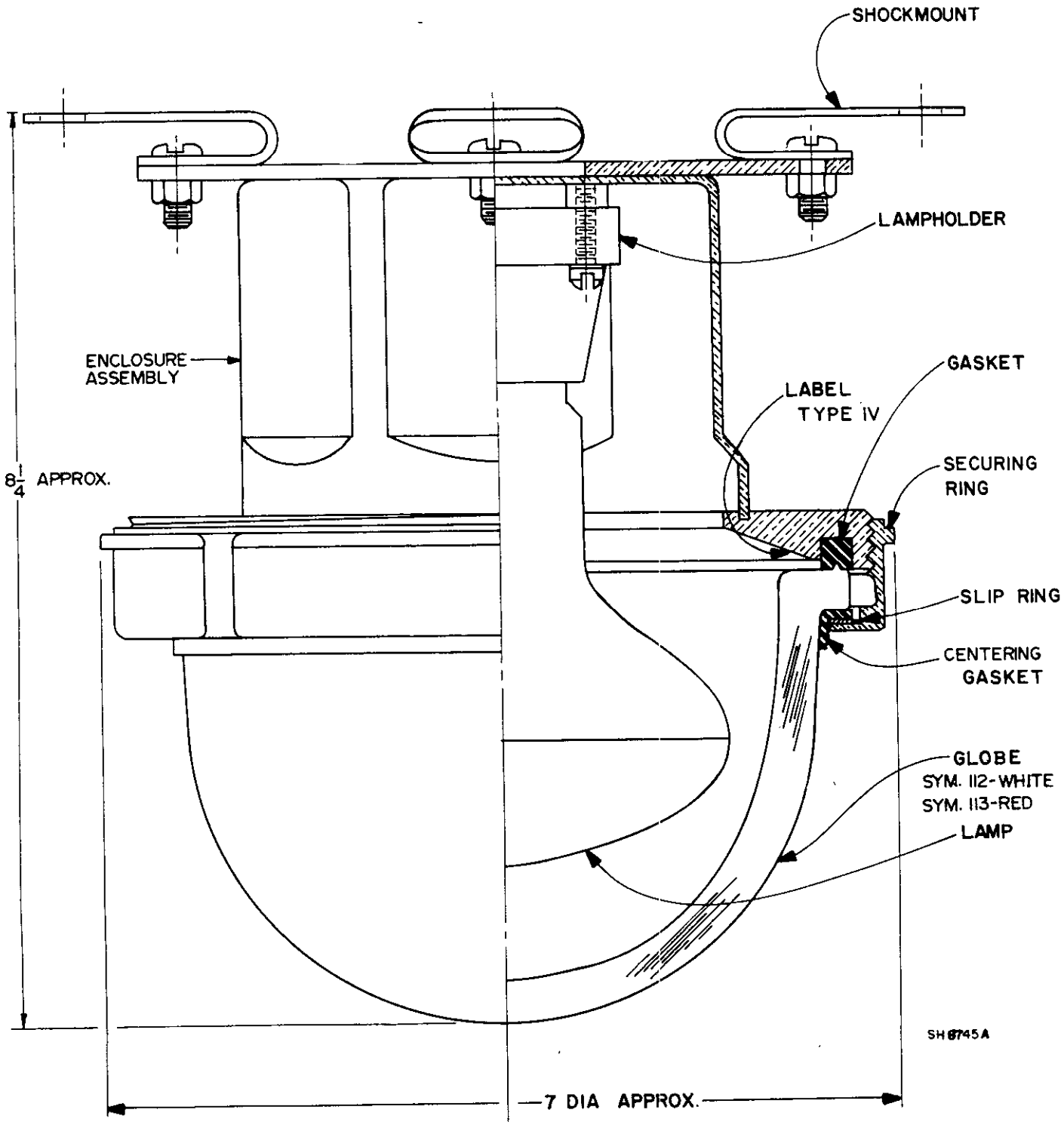


FIGURE 2. Fixture assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.

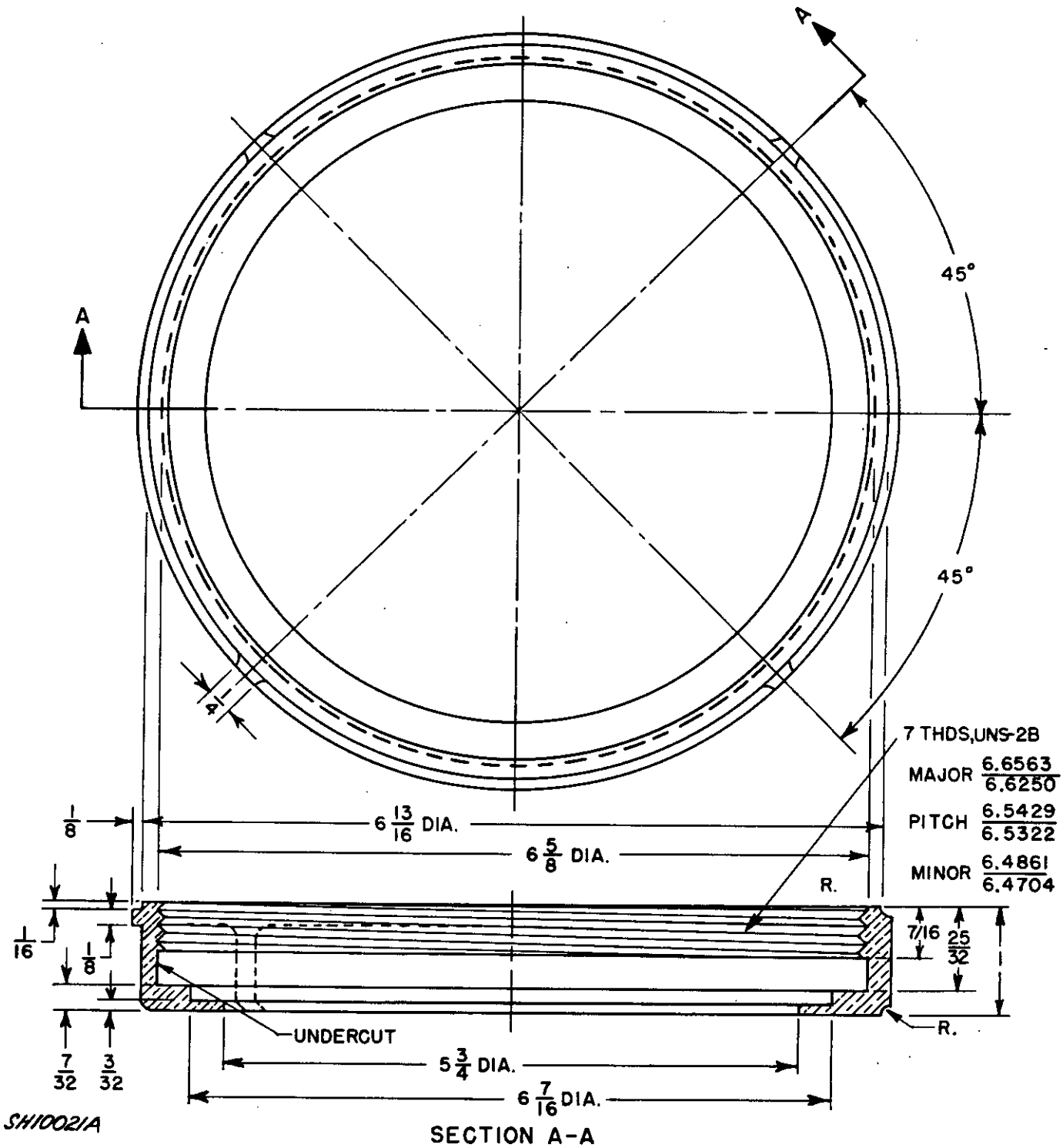


FIGURE 3. Securing ring.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.

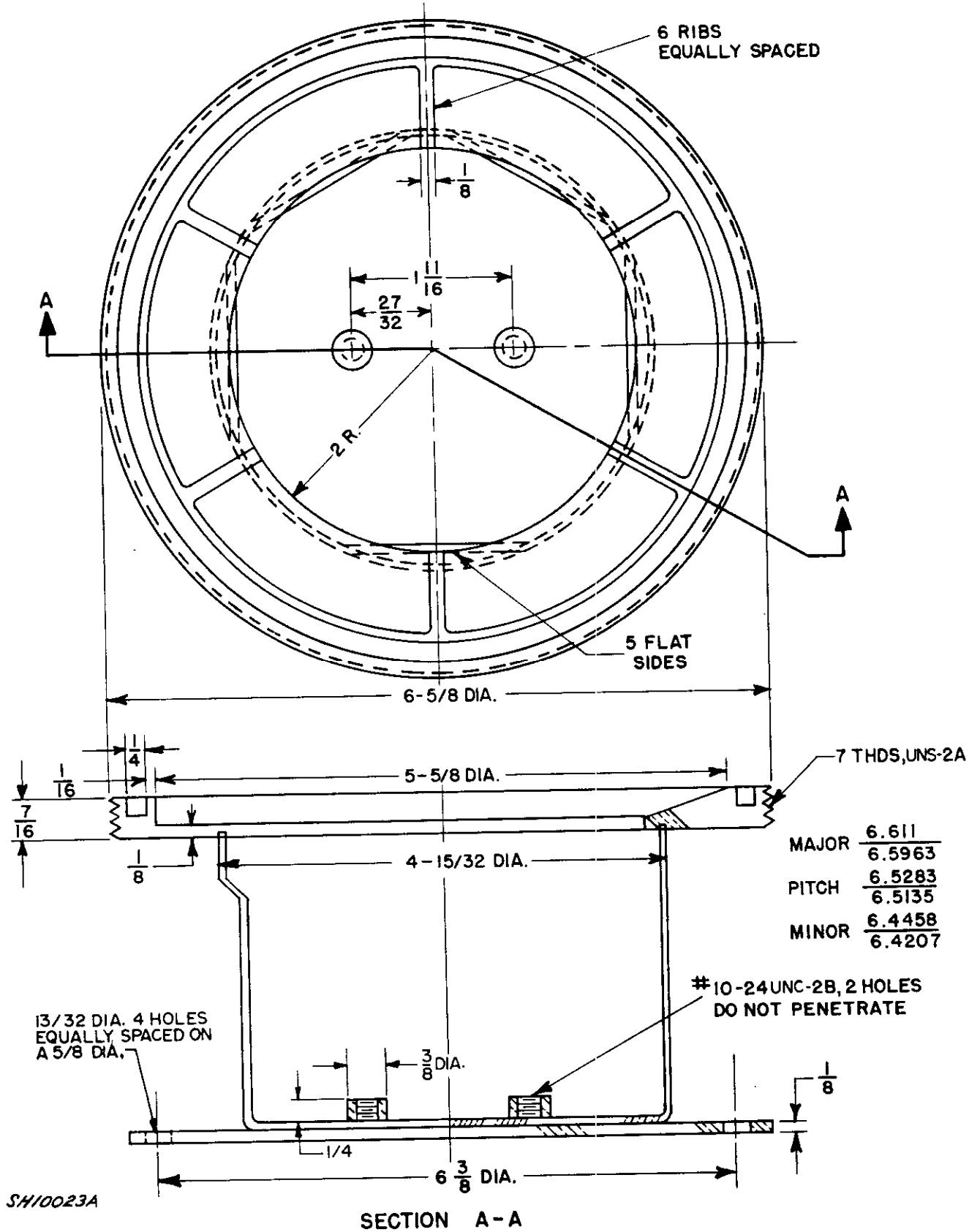


FIGURE 4. Enclosure assembly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.

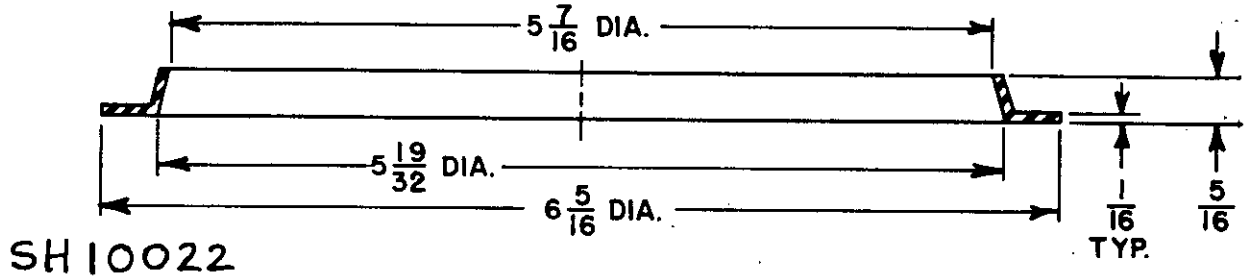


FIGURE 5. Globe centering gasket.

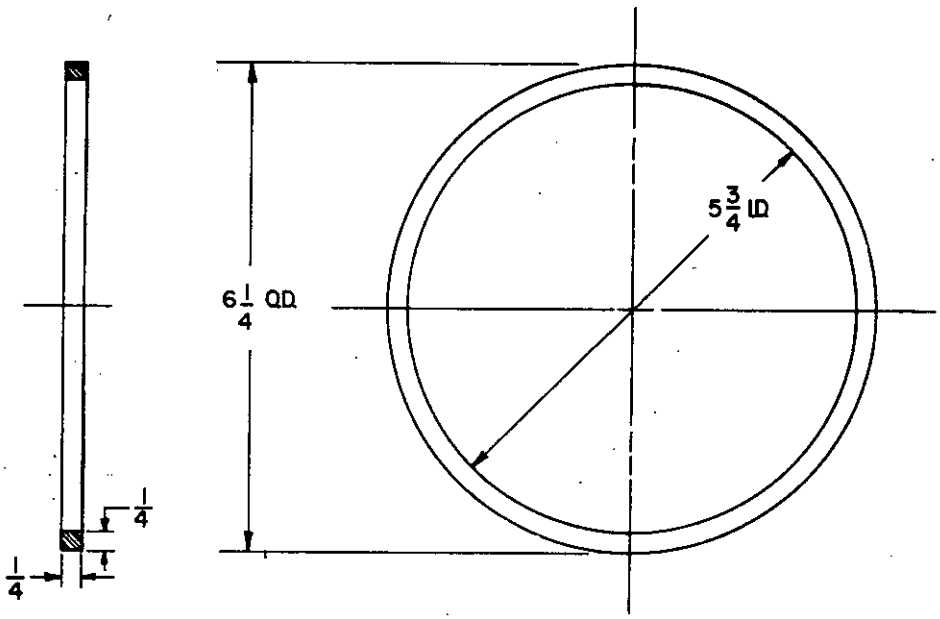


FIGURE 6. Enclosure gasket.

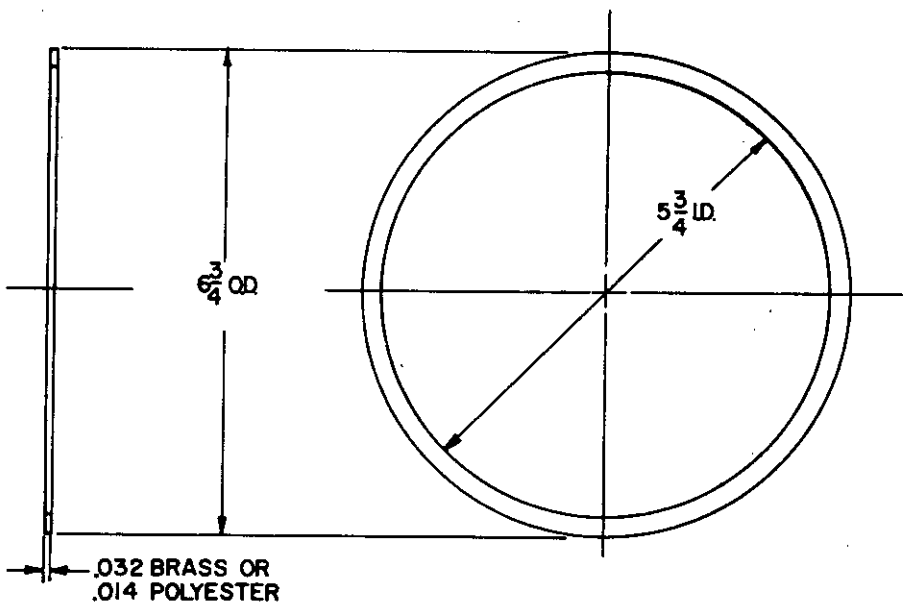
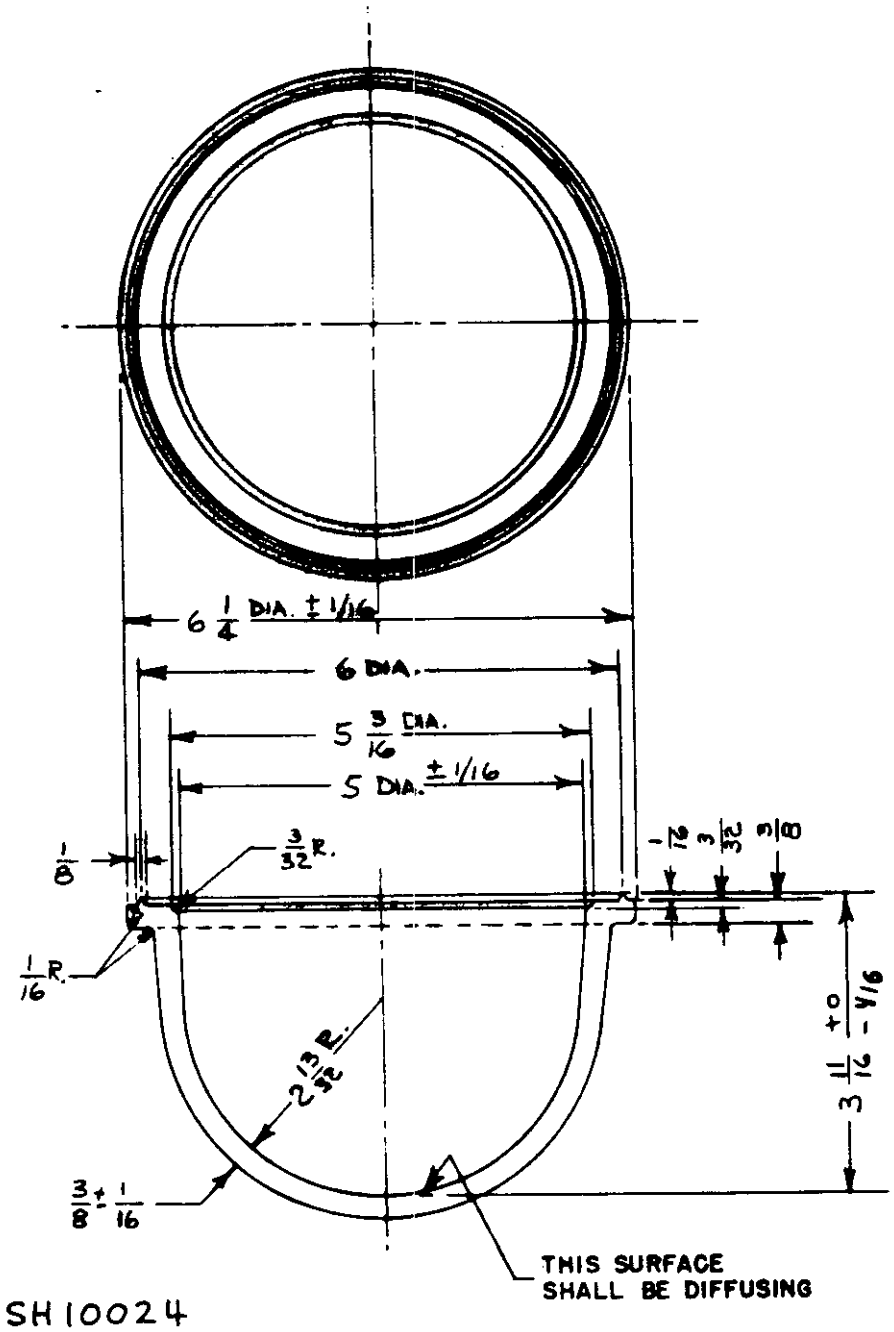


FIGURE 7. Slip ring.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.



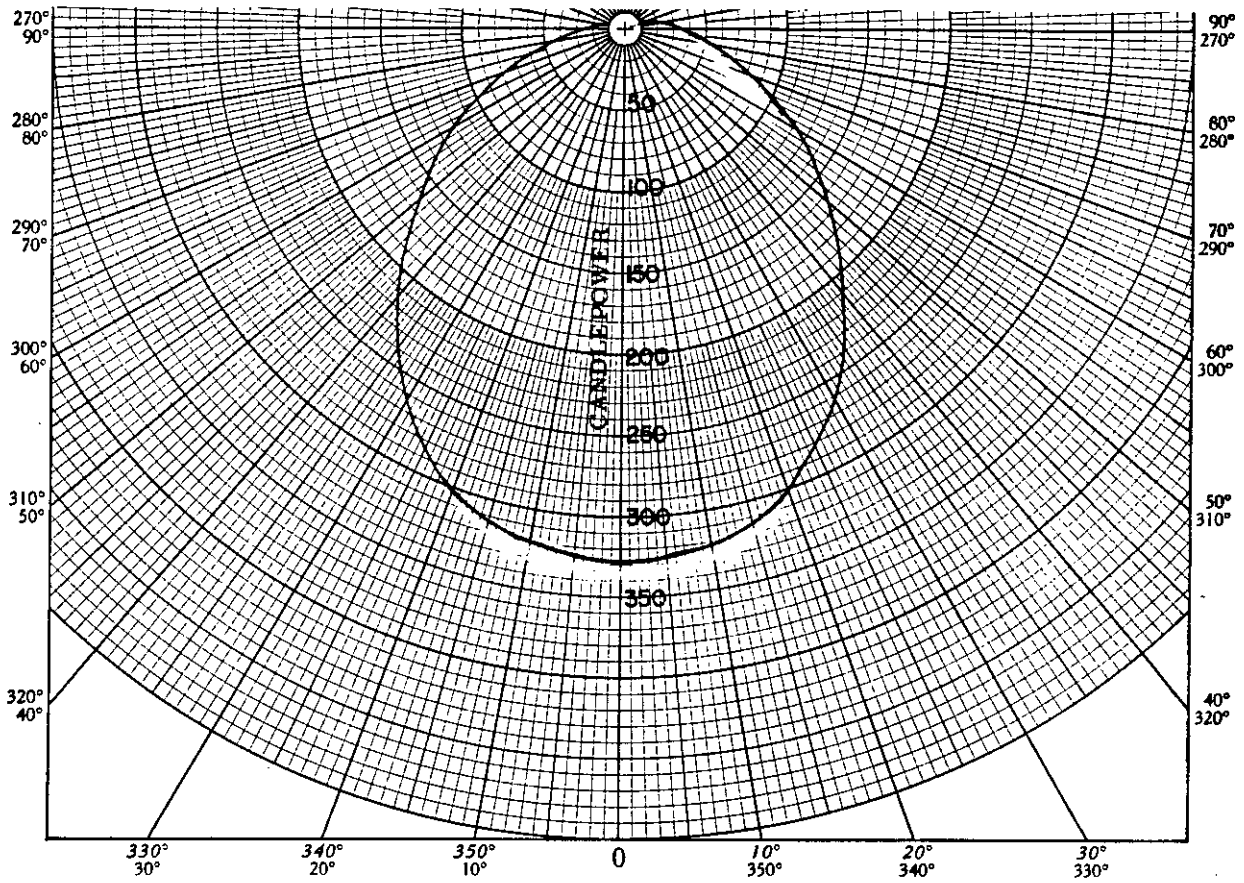
Material - High-impact, heat-resistant glass.
Color:

Symbol 112 - White (clear) transparent/diffusing.
Symbol 113 - Red transparent/diffusing.
Symbol 113.1 - Yellow transparent/diffusing.

Transmittance (symbol 112) - Clear transparent/diffusing - 80 percent minimum

FIGURE 8. Globe.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.



ANGLE	CANDLEPOWER SYMBOL 112
0	330
5	320
10	318
15	315
20	300
25	280
30	260
35	240
40	214
45	185
50	163
55	145
60	125

ANGLE	CANDLEPOWER SYMBOL 112
65	105
70	85
75	55
80	50
85	40
90	35
95	25
100	20
105	15
110	11
115	9
120	7
125	6

Photometric test data:

Distance: 10 feet.
Voltage: 120 volts.
Ambient temperature: 76°F.
Lamp: 110 watts, rough service conforming to MS16563-1, furnished by laboratory for test, shall be used.

FIGURE 9. Candlepower distribution curves.

Source: https://assist.dla.mil - Downloaded: 2026-10-16 08:45Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:
Type: II
Class: 2
Symbol 112: Fixture, general lighting, with one 110-watt white lamp and white (clear transparent) diffusing globe.
Symbol 113: Fixture, general lighting, with one 110-watt white lamp and red transparent diffusing globe.

Symbol 113.1: Fixture, general lighting, with one 110-watt white lamp and yellow transparent diffusing globe.

Design:
Electrical rating: 120 volts, 110 watts.
Dimensions and configuration: See figures 1 through 9.
Enclosure assembly: Brass figure 4.

Gaskets: Rubber, 1/4 inch square cross section by 5-3/4 I.D. by 6-1/4 O.d.
Lampholder: MIL-L-970/6, two contact.
Shockmounts: MIL-F-16377/41-007.
Globes:
Symbol 112: White (clear transparent) figure 8.
Symbol 113: Red, figure 8.
Centering gasket: Molded rubber figure 5.
Slip ring: Brass, 6-1/4 O.D. by 5-3/4 I.D., by 1/32 inch thick.
Securing ring: Cast brass or bronze, figure 3.
Lamp: This fixture shall be designed to accommodate a 110-watt, white lamp in accordance with MS16563-1. (Lamp shall not be furnished).
Weight (Symbols 112 and 113): 8 pounds, 9 ounces max.
Distribution curves (Symbol 112): See figure 9.
Label: Labels shall be furnished in accordance with MIL-F-16377. One type IV label shall be applied as shown on figure 2. The information to be shown on the type IV label shall be

"USE 110-WATT LAMP ONLY
LAMP: NSN6240-00-578-6820
WHITE GLOBE: NSN6210-00-635-8819
RED GLOBE: NSN6210-00-635-8820

QUALITY ASSURANCE: Quality assurance shall be as specified in MIL-F-16377 and as follows:

First article inspection:	
Complete fixtures:	See table I.
Globes.	See table II.
Quality conformance:	
Comparison inspection:	Same as the first article inspection.
Inspection of product for delivery:	
Examination:	See MIL-F-16377.
Examination for preparation of delivery:	See MIL-F-16377.
Quality conformance tests:	See MIL-F-16377.

TABLE I. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	
Insulation resistance	3.6.16	4.8.3	
Continuity of grounding	3.6.13.4	4.8.12	
Optical uniformity	3.7.5.1	4.8.4	
Mechanical shock	3.7.5.6	4.8.17.3	8-ft-lbs

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.

TABLE I. First article and comparison inspection for complete fixtures (Cont'd).

Tests	Requirement paragraph	Test paragraph	Remarks
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 113 only
Light output	3.7.1	4.8.6.1	Symbol 112 only Candlepower
Magnetic permeability	3.4.8	4.8.18	

TABLE II. First article and comparison inspection for globes.

Tests	Requirement paragraph	Test paragraph	Remarks
Examination	3.13	4.5	
Optical uniformity	3.7.5.1	4.8.4	
Shock	3.5.12	4.8.7	
Vibration	3.5.13	4.8.8	
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Light output	3.7.1	4.8.6.1	Symbol 112 only Candlepower
Mechanical shock	3.7.5.6	4.8.17.3	8 ft-lbs
Thermal shock	3.7.5.7	4.8.17.4	
Breakage pattern	3.7.5.9	4.8.17.6	
Color and luminous transmittance	3.7.3.1	4.8.6.3	Symbol 113 only

General information: (a) Assignment of Military part numbers and general information is shown in table III.

TABLE III. Part number and general information.

Military part number M16377/26-	Item name	National Stock Number (NSN)	Supersedes part number	Remarks
112	Complete fixture, Symbol 112 consisting of: Body Clear transparent diffusing globe	6210-00-635-8817 6210-00-635-8819		
113	Complete fixture, Symbol 113 consisting of: Body Red transparent diffusing globe	6210-00-635-8817 6210-00-635-8820		
113.1	Complete fixture, Symbol 113.1 consisting of: Body Yellow transparent diffusing globe	6210-00-635-8817		
001	Body: Complete fixtureless globe	6210-00-635-8817		
002	White (Clear transparent diffusing) globe	6210-00-635-8819	MS17802-10	
003	Red (transparent diffusing) globe	6210-00-635-8820	MS17802-11	
004	Securing ring	6210-00-399-6208	9000-S6401-74219-1	Figure 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T08:45Z
Check the source to verify that this is the current version before use.